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Forecasting Electricity Price Volatility for EV Charging Operations in the SP15 Market

ORSOL Forecasting

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ABSTRACT

Wholesale electricity price volatility creates operational risk for EV fast charging operators, particularly in California where renewable generation drives strong daily price swings. This study develops a 24 hour ahead forecasting framework for the SP15 zone using multi-year hourly data on market prices, load, renewables, weather, and fuel costs. A classical ARMA time series model is implemented in SAS to provide a transparent statistical benchmark, supported by SAS diagnostics and visualizations, while a leakage-controlled LSTM model is developed in Python to capture nonlinear patterns across 24-hour input windows. Together, these approaches identify predictable daily structure and highlight the limits of forecasting evening ramp spikes, offering practical insight for next day pricing and load shifting decisions in renewable driven markets and supporting more efficient integration of clean energy into EV charging operations.

INTRODUCTION

In California, EV charging operators increasingly rely on wholesale electricity markets rather than fixed retail rates. Rapid growth in renewable generation introduces strong intraday price patterns, particularly in the SP15 pricing zone, where the "duck curve" drives midday price declines and sharp early evening increases as solar output falls and net demand rises. These ramp hours are operationally critical because they coincide with commuter charging demand and the largest price swings of the day.

This study, conducted within Purdue University's VIP-ORSOL initiative, develops a 24 hour ahead forecasting framework for SP15 real-time electricity prices to support day-ahead pricing, load shifting, and risk management. We construct a multi-source dataset and implement both a classical ARMA benchmark in SAS, supported by diagnostic visualizations, and a leakage-controlled sequence-based forecasting pipeline in Python. The objective is to produce technically rigorous yet accessible results that inform operational decision making under renewable-driven market dynamics.

DATA

Our modeling dataset integrates multiple sources that influence or contextualize hourly electricity prices in Southern California. CAISO OASIS provides real-time locational marginal prices and related grid fundamentals such as system load and renewable generation forecasts. Weather variables, including temperature, wind speed, and solar irradiance, were drawn from historical datasets (e.g., NOAA and/or MERRA-2) and mapped to California pricing zones to represent demand drivers and renewable availability. Natural gas price

series (e.g., Henry Hub, PG&E Citygate, SoCal Border) were included because fuel costs can influence marginal generation economics and broader price conditions. In addition to these energy fundamentals, EV charging behavior was informed by session-level charging records (10,000+ sessions) from Purdue's DESL database, and charging station attributes (location, power ratings, port counts) were sourced from OpenChargeMap to support corridor-level operational interpretation.

After alignment and feature engineering, the primary modeling file (Forecasting_Final_Dataset_Enhanced.csv) contains approximately 28,000 hourly observations and 51 features. The target variable is LMP_RTM_SP15, the CAISO real-time market price for the SP15 zone in $/MWh. Predictors include load variables, renewable generation forecasts, weather by zone, natural gas hub prices, lagged price variables (e.g., lag_1, lag_2, lag_20), and engineered signals such as net load (Load − Solar − Wind), time encodings, and indicators for weekends and peak hours. The dataset is organized as an hourly multivariate time series designed to support 24-hour input windows for next-day prediction.

PROBLEM STATEMENT

The objective of this study is to predict SP15 real-time electricity prices 24 hours into the future. Formally, given a multivariate history of the prior 24 hours, the model produces a single forecast for hour $t + 24$. This horizon is operationally meaningful because charging operators need advance guidance to plan next-day pricing schedules, manage site utilization, and reduce exposure to predictable high-cost periods.

The problem is difficult because the SP15 price process is both structured and volatile. Intraday patterns driven by solar generation create predictable midday behavior, while the evening ramp between approximately 17:00 and 21:00 is marked by sharp transitions and frequent price spikes that are harder to anticipate. Extreme events (often > $100/MWh) occur during constrained supply or grid stress conditions, and longer-term seasonal dynamics introduce non-stationarity. Additionally, electricity datasets are prone to leakage if features inadvertently encode future market information. For example, day-ahead prices can be highly correlated with real-time prices and must be excluded when the goal is a fair, realistic forecast using only information available at prediction time. A persistence baseline—predicting the t+24 price as the value from 24 hours earlier—serves as a minimum standard that any useful model should exceed.

DATA CLEANING AND VALIDATION

All data cleaning, feature engineering, leakage prevention, and sequence construction for machine learning were implemented through a Python-based pipeline (including automated verification checks) to ensure realism and reproducibility. Raw time series were first standardized to a consistent hourly index and validated for continuity and ordering to prevent accidental look-ahead. Features were engineered with careful boundary control, including lag variables and rolling calculations designed to reference only past values. The target distribution was right-skewed due to price spikes, so a shifted log transformation was applied to stabilize variance and reduce sensitivity to extremes while preserving relative movements.

To prevent data leakage, the dataset was split strictly chronologically into training, validation, and test partitions (60%/20%/20%) with no shuffling, and checks were applied to confirm there was no temporal overlap. Scaling was performed in a leakage-safe manner by fitting normalization parameters only on the training split and applying them unchanged

to validation and test data. A correlation-based screening step was used to identify and remove features that were overly predictive due to market leakage pathways (including day-ahead SP15 prices and other highly correlated market components). Additional safeguards verified that lag features were correctly constructed, that no sequences crossed split boundaries during window creation, and that each training example used only historical information.

SAS was used specifically for the classical time-series benchmark stage: an ARMA model was trained and evaluated in SAS to provide an interpretable baseline for short-term forecasting performance. SAS diagnostic outputs and detailed plots were used to assess residual structure, stability, and fit quality, offering a transparent point of comparison against sequence-based machine learning approaches. This division of labor supports both rigor and accessibility: Python enforces strict pipeline realism for model training, while SAS supports benchmark modeling and diagnostic visualization that clearly communicates model behavior to a broad audience.

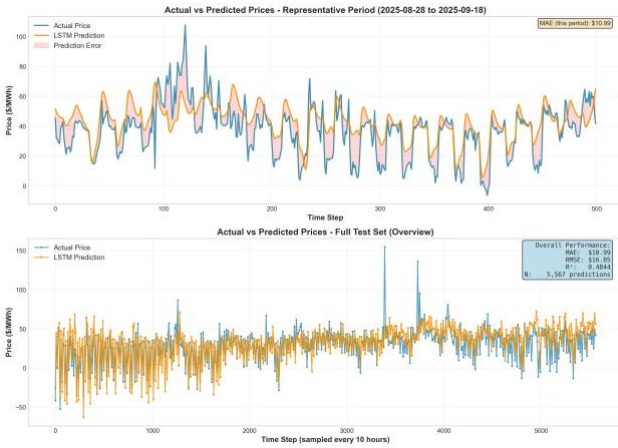
ANALYSIS

The LSTM model was designed to capture nonlinear temporal patterns in SP15 real-time electricity prices using 24-hour rolling input windows. Following a strict chronological 60/20/20 split to prevent leakage, feature selection and scaling were performed on training data only, with price-related variables removed to ensure realistic forecasting. Each sample used 24 hours of multivariate inputs to predict the price 24 hours ahead. The stacked LSTM architecture (128→64→32 units with dropout) learns both short-term persistence and cross-variable dependencies. Trained with Huber loss, early stopping, and adaptive learning rate reduction, the model achieved an out-of-sample MAE of about $10.99 and R^2 of roughly 0.48 on ~5,500 test observations. It tracks daily structure well but shows higher error during evening ramp hours (17–21), where volatility is greatest.

The ARMA benchmark was implemented in SAS Viya to evaluate how much of SP15 real-time price variation can be explained through linear temporal structure. Using SAS time series procedures, we estimated an ARIMA(2,0,1) model with seasonal and hourly regressors, followed by a SARIMA(2,1,1)(1,1,1)[24] specification to explicitly model the 24-hour cycle confirmed through ACF and PACF analysis. An extended ARIMAX model incorporating temperature, solar generation, and actual load was also tested to account for physical drivers of price formation. All models were estimated using Maximum Likelihood Estimation and evaluated using AIC and SBC criteria. Results confirm strong short-term persistence, particularly dependence on the prior hour and prior day, and seasonal visualizations generated with SAS SGPLOT show stable baseline behavior across years. However, residual diagnostics indicate that extreme price spike events remain poorly captured across specifications, reflecting the limitations of linear models in volatile electricity markets. These results establish ARMA-based modeling in SAS Viya as a transparent and interpretable statistical baseline for short-term forecasting.

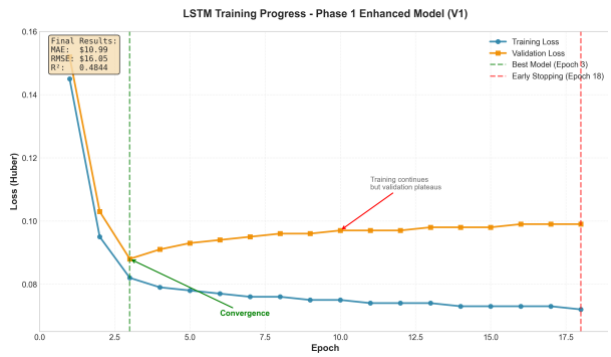
VISUALIZATION

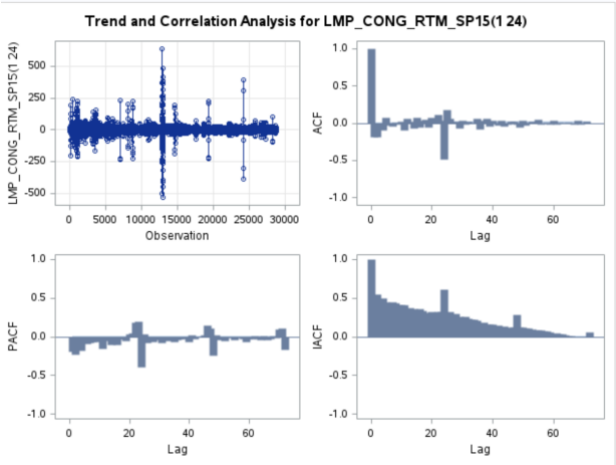
Five visualizations were used to evaluate model behavior and communicate results clearly.



The top panel displays a 500-hour representative window (Aug 28 – Sep 18, 2025) where model MAE closely matches overall test performance (~$10.99). Actual prices (blue) and 24-hour-ahead LSTM predictions (orange) are shown, with shaded regions highlighting forecast error. The model tracks daily price structure, including midday troughs and evening ramps, but underestimates sudden spikes. The bottom panel summarizes the full test set (~5,567 forecasts), sampled for readability, and reports MAE (~$10.99), RMSE (~$16.05), and R^2 (~0.48), confirming consistent performance across the evaluation period.

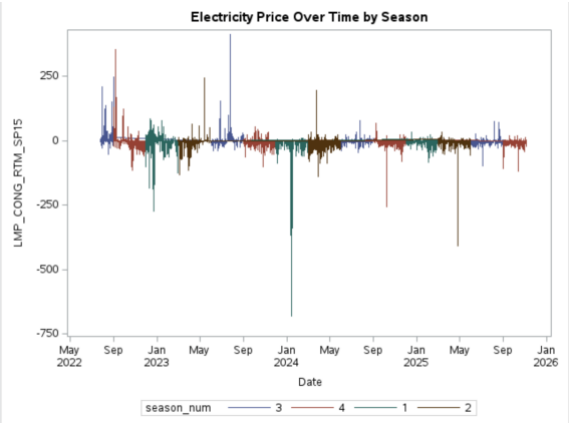
The loss curve shows Huber loss over epochs. Validation loss reaches its minimum at Epoch 3, after which improvement plateaus while training loss continues to decline. Early stopping at Epoch 18 prevents overfitting. The rapid convergence indicates that the model captures most predictable structure early while maintaining generalization.





The 24-hour fluctuation plot illustrates strong temporal persistence, with prices closely linked to the prior hour and the same hour on the previous day. This supports the autoregressive structure of the SAS ARMA benchmark model.

Seasonal comparison shows relatively stable baseline prices across years, with volatility spikes occurring in all seasons. This suggests that extreme price events are driven more by grid stress and market conditions than by seasonal shifts alone.



SUGGESTIONS FOR FUTURE STUDIES

Future work could extend this framework in several directions. First, incorporating probabilistic forecasting methods would allow operators to quantify uncertainty during high-volatility ramp hours rather than relying solely on point predictions. Second, integrating battery storage behavior or demand-response participation into the modeling framework could better reflect real operational flexibility at charging sites. Third, expanding the analysis to additional CAISO pricing zones would test model generalizability across different renewable penetration levels and grid conditions. Finally, combining price forecasts with real charging demand models could enable joint optimization of pricing and load scheduling, moving from prediction toward fully integrated operational decision support.

DISCUSSION

The results show that a meaningful portion of SP15 real-time electricity price behavior is predictable 24 hours in advance, particularly the recurring intraday structure driven by load and renewable generation patterns. The SAS-based ARMA model captures linear persistence effectively, while the LSTM improves performance by modeling nonlinear relationships across multiple inputs, achieving an out-of-sample MAE of approximately $10.99 and explaining nearly half of observed variance ($R^2 \approx 0.48$). Both models track daily troughs and ramps well but struggle with sudden evening volatility and extreme price spikes, especially during peak ramp hours. These findings suggest that while rare grid stress events remain difficult to anticipate, structured price dynamics can be leveraged for operational planning and risk reduction.

CONCLUSION

This study develops a 24-hour ahead forecasting framework for SP15 real-time electricity prices to support EV charging operations along the I-210 corridor. By combining a SAS-implemented ARMA benchmark with a leakage-controlled LSTM model, we demonstrate that nearly half of short-term price variability can be explained using historical market, load, weather, and fuel data. While abrupt volatility events remain challenging, the models provide actionable insight for next-day pricing and load-shifting decisions. The results highlight the value of integrating classical time-series methods and machine learning within a unified analytical workflow to support efficient, data-driven EV infrastructure planning and more sustainable alignment of charging demand with renewable energy availability.